Name_____ Date_____ Period_____

LESSON 7-3 · HOMEWORK PACKET · SHEET 1 OF 2

Absolute Value as Distance

Maryland Standard 6.NS.C.7c · Learning target – I can use absolute value to describe how far a number is from zero.

REMEMBER

- **Absolute value** is a number's distance from 0. We write $|-7| = 7$.
- A number and its opposite have the same absolute value: $|5| = |-5| = 5$.
- Absolute value is never negative. A greater absolute value means **farther from 0**.

WORKED EXAMPLE

Which is farther from 0: −9 or 6?

Step 1 $|-9| = 9$ and $|6| = 6$.

Step 2 $9 > 6$, so −9 is farther from 0.

Answer **−9**

PRACTICE

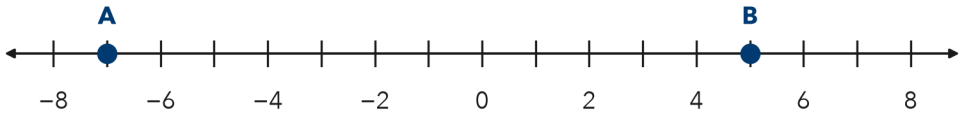
show your work · label units

1. Find each absolute value.

a. $|-14|$ = _____ **b.** $|8|$ = _____ **c.** $|0|$ = _____

d. $|-2.5|$ = _____ **e.** $|3/4|$ = _____ **f.** $|-1\ 1/2|$ = _____

2. Use the number line. Find the absolute value of each point.



a. $|A|$ = _____ **b.** $|B|$ = _____ **c.** Farther from 0: ____

3. Compare. Write <, >, or = in each blank.

a. $|-9|$ ___ $|7|$ **b.** $|-4|$ ___ $|4|$ **c.** -10 ___ $|-3|$

d. $|-6.5|$ ___ $|6.5|$ **e.** $|-12|$ ___ $|-15|$ **f.** 0 ___ $|-1|$

4. **Bank Balances.** Rosa's account balance is −$35 (she owes $35). Ben's balance is $20.

a. Find the absolute value of each balance. _____

b. Whose balance is farther from $0? _____

5. **Stretch.** List all the integers with an absolute value less than 3.

Integers_____

6. **Reasoning. Name two different numbers that have an absolute value of 4.5. Explain how you know. Sentence Frame: "I know this because..."**

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask someone at home: is it colder at −10 degrees or at 5 degrees? Which temperature is farther from 0?

Name_____ Date_____ Period_____

LESSON 7-3 · HOMEWORK PACKET · SHEET 2 OF 2

Absolute Value as Total Distance

Maryland Standard 6.NS.C.7c · Learning target – I can use absolute value to find the total distance between a negative and a positive reading.

REMEMBER

- **Opposite sides** of 0: **add** the absolute values.
- −3 to 5 is 3 + 5 = 8 units.
- **Same side** of 0: **subtract** the absolute values.
- −9 to −2 is 9 − 2 = 7 units.

WORKED EXAMPLE

A diver swims up from −24 feet to a boat deck at 6 feet. How far does she go?

Step 1 The numbers are on opposite sides of 0, so add.

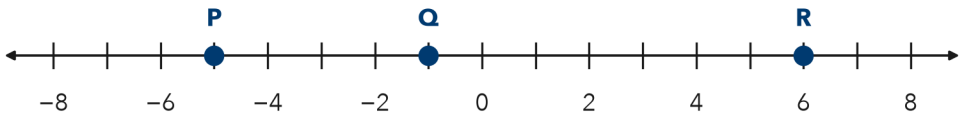
Step 2 |−24| + |6| = 24 + 6 = 30

Answer **30 feet**

PRACTICE

show your work · label units

1. Use the number line. Find each distance.



a. P to R: _____ **b.** Q to R: _____ **c.** P to Q: _____

2. Find the distance between each pair of numbers.

a. −7 and 3 _____ **b.** −12 and −4 _____ **c.** −2.5 and 4 _____

3. **Temperature Swing.** It was −8 degrees at 6 a.m. and 15 degrees at 3 p.m. How much did it rise?

Answer_____ degrees

4. **Elevator.** An elevator goes from parking level −3 up to floor 9. Street level is floor 0. How many floors does it travel?

Answer_____ floors

5. **Find the Error.** To find the distance from −9 to 5, Omar wrote 9 − 5 = 4. Explain his mistake. Find the correct distance.

_____

_____

TEST PRACTICE

choose the best answer

6. What is the distance between −13 and 6 on a number line?

A 7 **B** 19 **C** −19 **D** −7

7. A cave floor is at −60 feet. A ledge is at −22 feet. How far apart are they?

A 82 feet **B** 38 feet **C** −38 feet **D** 22 feet

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME With someone at home, find how many degrees the temperature changed today.